



Experimental Investigation of a Skin-Actuated-Camber (SAC) Morphing Wing Design

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The aerodynamic performance of a novel Fishbone Skin-Actuated-Camber (SAC) morphing wing design, which actuates its skin to change its effective camber, was studied both experimentally and numerically. Force-based experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) to compare the performance of four morphing wing designs with different hinge locations, two ideal trailing edge flap wings, and one conventional trailing edge flap wing. All test articles have an Eppler 479 airfoil, an effective aspect ratio of four, and were tested within an angle of attack range of -15° and 15° . The novel design achieved effective camber change without any buckling, maintaining comparable aerodynamic performance to ideal flap wings at a Reynolds number of 270,000. At a Reynolds number of 400,000, the morphing shows a lower drag than the ideal flap wing. Simulations from FlightStream®, a numerical solver correlated well with experimental lift data, with the morphing wing's pressure contours indicating reduced flow separation and gradual pressure change on the upper surface when deflected.

Nomenclature

AR	= Aspect Ratio	D	= Drag force, (N)
c	= Normalized Chord	L	= Lift force, (N)
c_{def}	= Chord Location of Deflection	q	= Dynamic Pressure, (Pa)
C_D	= Drag Coefficient, $C_D = \frac{D}{qS_{Ref}}$	Re	= Reynolds Number
C_{D_0}	= Zero Lift Drag Coefficient	S_{ref}	= Reference area, (m^2)
C_L	= Lift Coefficient, $C_L = \frac{L}{qS_{Ref}}$	V_∞	= Freestream Velocity, (m/s)
C_p	= Pressure Coefficient	α	= Angle of Attack, (Deg)
C_{Di}	= Induced Drag Coefficient	ρ	= Air density, (kg/m^3)
		δ_{eff}	= Effective Camber Angle, (deg)

I. Introduction

The concept of the morphing wing which includes wing twisting, wing sweeping, wing folding, span extending, and camber changing has been around for decades [1]. Recently, the interest in camber-changing morphing wings has increased with the advent of fixed-wing Unmanned Aerial Vehicles (UAV). Traditional trailing edge hinged devices cause significant drag penalties due to gaps between the wing body and the trailing edge device. An ideal morphing wing has a smooth surface transition and continuous change in camber providing better aerodynamic performance. Many morphing wing designs have been investigated analytically, experimentally, and computationally since the 1980s [2,4,7,11-14]. Applying the morphing wing concept to modern fixed-wing UAVs could provide better aerodynamic performance, thus extending the range for drone operations.

To achieve effective camber change, the wing airfoil shape can be morphed either at the leading edge or the trailing edge. Sodja et al. [3] explored the structural performance of a morphing leading-edge design, facing challenges due

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to the small radius at the wing's leading edge resulting in skin buckling. In morphing wings, trailing edge morphing is more common than the leading edge. Henry et al. [4] designed and analyzed the structural performance of a morphing wing using MFC piezoelectric patches, which provided a smooth morphing structure and a better overall stiffness. Bilgen et al. [5] constructed a 1.04m wingspan UAV featuring the piezoelectric morphing wing. Although it achieved a sufficient roll rate, it was about ~45% lower than that of the traditional aileron actuation on the same aircraft. Vasista and Tong [6] designed a morphing structure with pressurized chord-wise cells, achieving morphing through the differential pressure on the airfoil's upper and lower side, causing the cell wall to buckle and deform. However, the changes in the camber line due to morphing were limited in the study.

Modern morphing wing designs are usually made of a combination of rigid and soft materials. Yokozeki and Sugiura [7], Schlup et al. [8], and Navaratne et al. [9] designed a morphing wing internal structure using corrugated shapes to avoid skin buckling. However, issues such as discontinuities at the wing's lower surface increased drag [7]. Pankonien et al. [10] designed and built a more complicated morphing wing using a macro-fiber composite where the trailing edge was actuated by metal wiring and a hinge system. A silicon skin was used to provide a smooth surface morphing during the actuation process. However, the durability of the silicon skin and the strength of the actuation mechanism were significant drawbacks. Mkhoyan et al. [11] created a glass fiber morphing wing with six individual morphing sections along the span, using linear actuators and silicon fillings to ensure smooth morphing.

Bio-inspired fish-bone morphing concepts have gained traction in recent years, especially with the study conducted by Rivero et al. [12-14]. Their first iteration had fish-bone shape stringers at 40% chord location, a solid bending beam spline, soft 'tendons' for actuation, and an elastomeric matrix composite skin. Even though the wing morphing was achieved, it lacked sufficient stiffness for wind tunnel testing. Their second iteration was improved by moving the morphing location to 75% chord and using more rigid materials, achieving a lift coefficient change of 0.7 and up to 70% drag reduction compared to traditional flaps, as shown by Particle Image Velocimetry (PIV) results.

Except for the piezoelectric morphing wing concept, none of the other morphing wing designs was able to achieve camber morphing using skin actuation and simple mechanisms that can be manufactured easily without using special materials. For a smaller-size fixed-wing UAV, a simpler mechanical and reliable morphing design that outperforms the conventional trailing edge device is desirable. The current study introduces a novel morphing wing design using simple mechanical skin actuation which can be rapidly prototyped using traditional 3D printing filaments. The product can be used for any small-scale fixed-wing UAVs.

II. Experimental setup

All experiments were conducted in the University of Dayton's Low Speed Wind Tunnel (UD-LSWT) in its open jet configuration. The tunnel has a 16:1 contraction ratio, six anti-turbulence screens, and a 0.762m by 0.762m open jet test section. The effective length of the test section is 0.183m. A 1.128m by 1.128m collector collects the expanded air onto the diffuser. A pitot tube is connected to a Siemens QBM-3230-U5D analog pressure transducer to measure the freestream velocity during the experiment. A TSI T600 Micromanometer was used for the pressure transducer calibration and a NI-USB-6259 DAQ was used to collect the pressure transducer data at the same sampling rate as the F/T transducer.

A. Experimental Setup

The force-based experimental setup is shown in Figure 1. The wing model was mounted to a rotary stage and an ATI Gamma F/T [14] balance assembly. The sensor's calibrated uncertainty was 1.25% for the Y-axis, which measures the normal force of the wing, and 0.75% for the X-axis, which measures the axial force of the wing. A 0.914m x 1.829m splitter plate enabled half-span model testing. The test matrix of the experiment is shown in Table 1. The freestream velocity for the morphing wing case was set at 20 m/s and 30 m/s, corresponding to a chord-based Reynolds number of 270,000 and 405,000 to investigate the Reynolds number effect on the morphing wing design. For $Re = 270,000$ cases, the angle of attack was varied from -15° to 15° with a 1° increment. While for the $Re = 405,000$ cases, the angle of attack was adjusted based on the maximum rolling moment the wing could achieve, due to the F/T sensor's torque limitations. Two trials were run in each case to ensure data repeatability. To minimize sensor drift, tare values with the wind tunnel powered off were taken before and after each test. The average of the two tares was subtracted from the respective thrust and torque readings. If the difference between the before and after tare value was greater than 0.1N of force on any axis, the experimental data was then rejected, and the experiments run were conducted again.

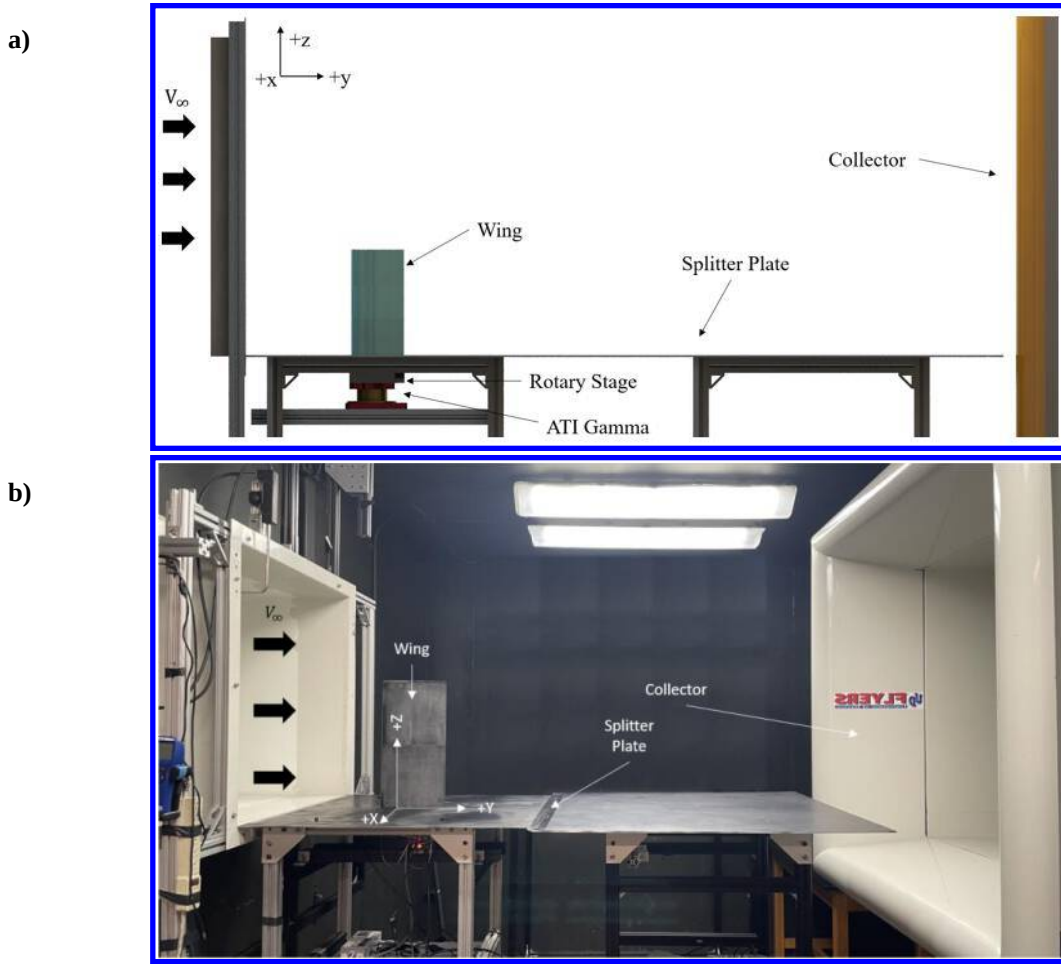


Figure 1. a) Rendered CAD Schematic and b) Picture of UD-LSWT Experimental Setup

Table 1. Test Matrix for Experimental Investigations

Wing Type	Reference Area(m²)	Reynolds Number	Angle of attack (deg)	Freestream Velocity (m/s)	Chord Location of Deflection	Effective Camber Deflection (deg)
Ideal Flap	0.0826	270,000 405,000	-15 to 15 -8 to 6	20 30	42.5%, 70%	0.00, 3.00, 5.00
Conventional Flap					70%	0.00, 2.50, 4.50
Morphing Wing					30%	0.00, 2.42, 3.85
					42.5%	0.00, 2.48, 4.45
					55%	0.00, 2.52, 4.75
					70%	0.00, 2.64, 4.98

B. Morphing Wing and Trailing Edge Flap Wing Structure

Four different patent pending Fishbone Skin-Actuated-Camber (SAC) morphing wings were designed with different chordwise morphing locations. All morphing wings were designed using the Eppler 479 airfoil, with a span of 0.406m, a chord of 0.203m, and a half-span aspect ratio (AR) of 4.0. Figure 2 shows the general design of all Fishbone SAC morphing wings using 70% chord location of deflection (c_{def}) model as an example. The wings feature a rigid leading edge and spar section made of PLA plastic, and a flexible trailing edge and skin section 3D printed

with a thin layer of PLA to maintain flexibility. Several design iterations are conducted to reduce the weight of the structure while preserving the same morphing capability. Four AGFRC B13DLM servo motors, located at the wing root, mid-span, and wingtip were used to actuate the morphing wing. An Arduino Uno microcontroller board was used to control these motors at three different deflection angles which will be discussed later in the section. Two end caps at the root and tip shield the internal structure from freestream and reduce drag.

Figure 3 shows the cross-sectional view of the four Fishbone SAC morphing wing designs, with morphing locations at 30%, 42.5%, 55%, and 70% chord. The interior airfoil section of the morphing location is rigid, while the outer skin is flexible. Servos near the leading edge of the wing, fitted with custom-made servo arms, are connected to the upper and lower surface skin section using a 2mm stainless steel rod along the span. A skin spine connects the rigid section to the trailing edge of the skin to maintain the basic airfoil shape of the morphing wing. A 'T-shape' latch mechanism between the spine and the skin helps maintain the airfoil shape while still allowing for deflection. For morphing locations closer to the trailing edge, a leading-edge latch mechanism performs the same function (shown in Figure 3b). For the $c_{def} = 0.7$ wing (Figure 3.d), the skin spine is removed to improve the morphing capability as the short flexible skin section is strong enough to hold its shape. The effective camber angle change (δ_{eff}) was calculated to compare camber changes quantitatively, measured from the centerline of the neutral position to the wing's leading-trailing edge line (Figure 3a). Notably, δ_{eff} slightly increases with c_{def} , meaning the same servo travel impacts wings with smaller flexible trailing edge sections more significantly.

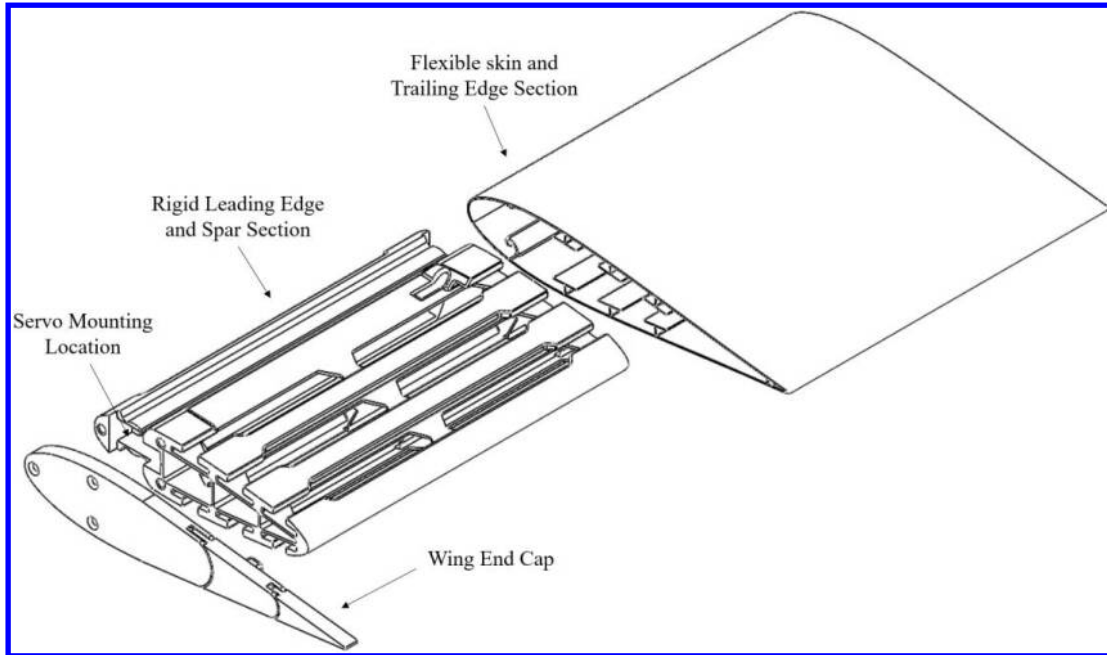


Figure 2. Schematic Showing the General Design of the Fishbone SAC Morphing Wing ($c_{def} = 0.7$) (Patent pending)

To further compare the performance of the morphing wing with a conventional flap design, the ideal Eppler 479 airfoil flap, which removes any servo, opening slots, or actual hinges, was tested. These ideal flap wings have the same planform area as the morphing wing. Two different hinge locations were selected corresponding to the morphing wing's c_{def} . The airfoil shapes for the trailing edge devices are shown in Figure 4. To vary the δ_{eff} for the ideal flap model, different swappable trailing edge pieces were installed for various flap deflection scenarios, as shown in Figure 4b. Three different δ_{eff} are selected and tested in the current study for the conventional trailing edge devices: 0.0° , 3.0° , and 5.0° . To further compare the morphing wing performance with real-world applications, an additional conventional flap with $c_{def} = 0.7$ is also designed, as shown in Figure 5 with hinges and servo actuation. The airfoil and surface area of the conventional trailing edge flap design are identical to those of all morphing wing designs. Four AGFRC B13DLM servo motors, the same as those used in the morphing wing design, were used to actuate the flap section of the wing with the connection rod in between. Two δ_{eff} (2.5° and 4.5°) were achieved with half and full servo deflection, closely matching the δ_{eff} of all morphing wing designs.

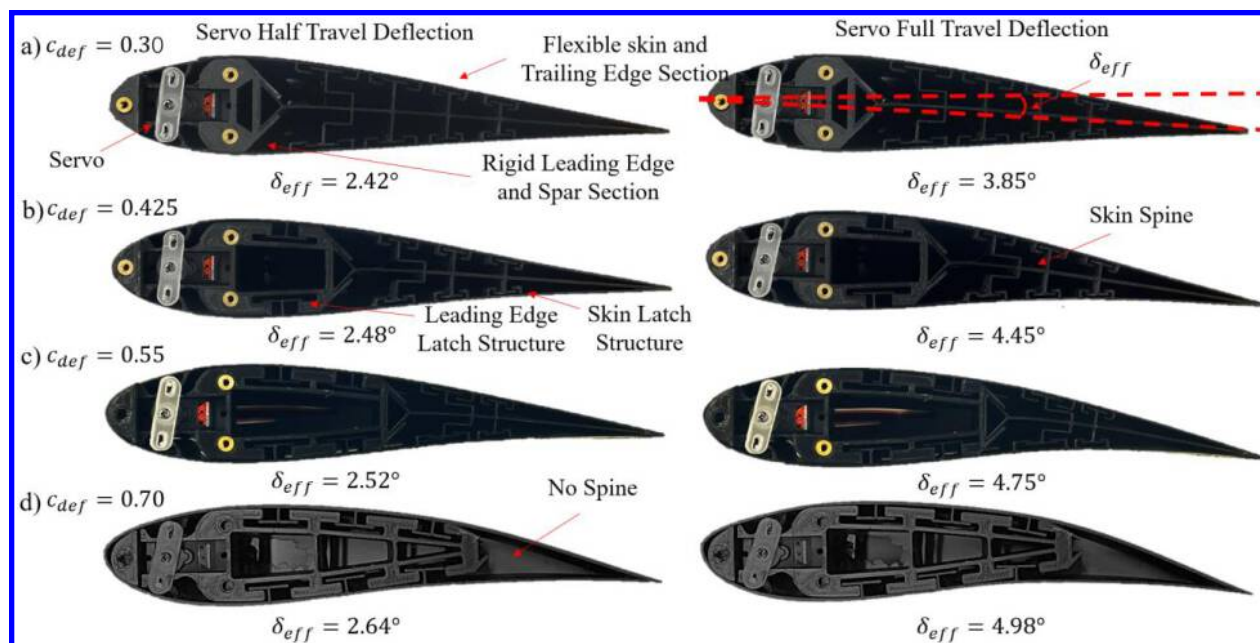


Figure 3. Cross-sectional View of the Morphing Wing with c_{def} of a) 30%, b) 42.5%, c) 55%, and d) 70% at Servo Half and Full Travel Deflection (Patent pending)

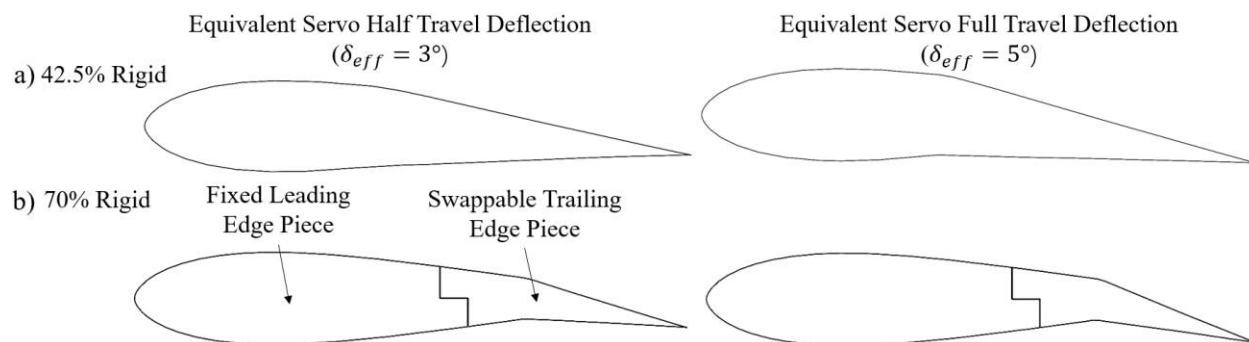


Figure 4 Schematic Showing the Idea Flap Airfoil Model with c_{def} of a) 42.5% and b) 70% Leading-Edge Section Under Different Equivalent Servo Deflections

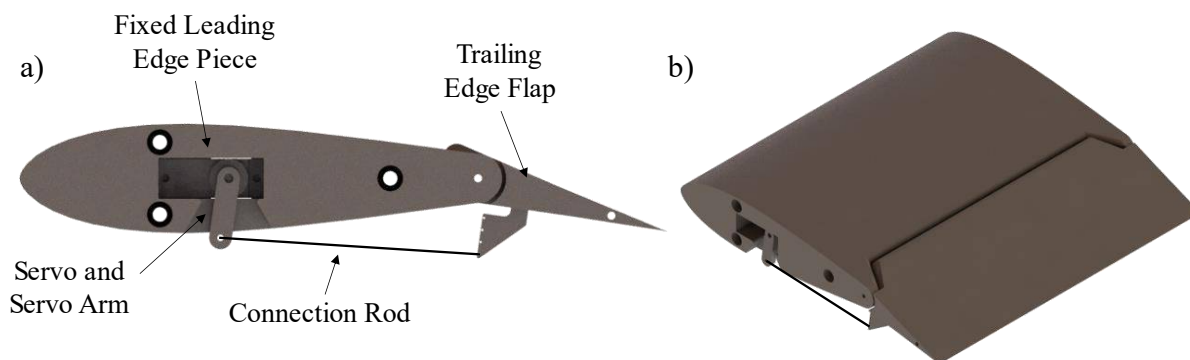


Figure 5. a) Side View and b) Isometric View Showing the Conventional Flap Model Commonly Used on Small Scale UAV.

C. Numerical Investigation with FlightStream®

FlightStream® [14] was used to numerically investigate the same models tested in the wind tunnel to obtain the pressure distribution comparison between the conventional and the morphing wing. FlightStream® contains two main types of potential flow solvers: pressure-based solver and vorticity solver. The pressure-based solver is used in the present study, which utilizes the pressure fields on the model geometry to determine the aerodynamic loads. A combination of source and sink panels along with doublet panels on the trailing edge is used to provide a solution for pressure-based potential flow solvers.

For the current study, the pressure-based solver is selected based on good agreement with the wind tunnel experimental result in our previous studies [17]. The boundary layer is assumed to be fully turbulent with the viscous coupling enabled. The surface mesh of the wing and the splitter plate in FlightStream® is shown in Figure 4. The wing platform has a structured mesh with 80 node points spanwise and 60 node points chordwise on both upper and lower surfaces. The chordwise mesh also has a dual side growth rate of 1.05 to better resolve the leading and trailing edge geometry. Moreover, the splitter plate of the same size as the one used wind tunnel experiment is included in the model to better represent the actual experimental conditions. The splitter plate has 60 node points from side to side and 80 node points, with a dual-side growth rate of 1.05 downstream.

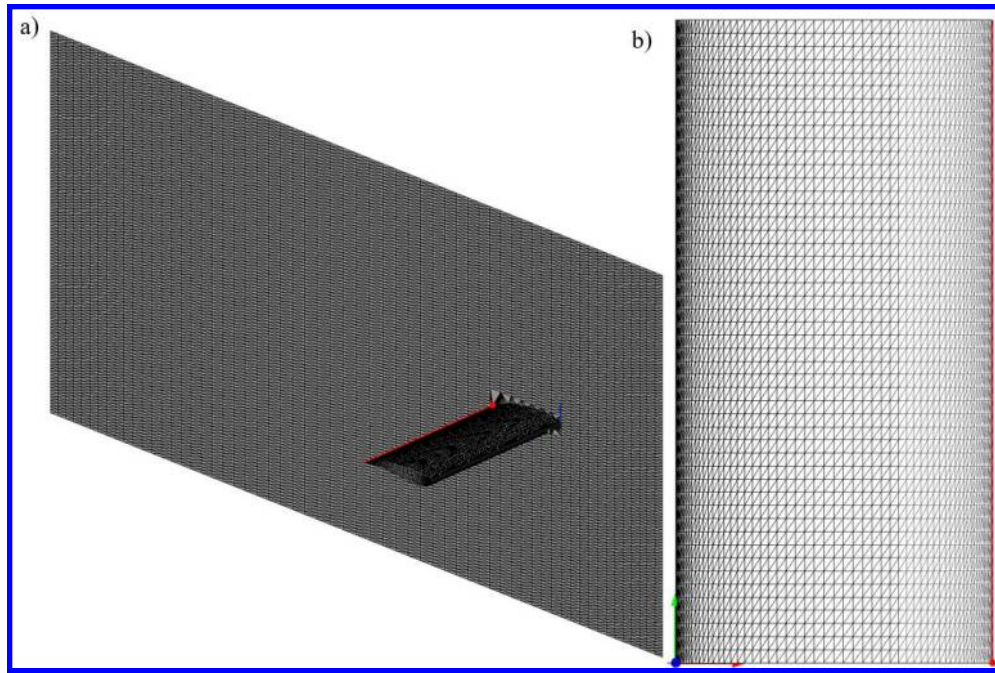


Figure 6. Surface Mesh in FlightStream® for a) Isometric View of the Wing and Splitter Plate and b) Top View of the Wing

III. Results

A. Experimental Results

The wind tunnel experimental results will be discussed in this section. Initially, the aerodynamics of the conventional hinged trailing edge flap wing and the ideal trailing edge flap wing at $Re = 270,000$ will be discussed first. These results will act as a baseline for comparing the morphing wing, which will be discussed subsequently. The performance of all four morphing wing designs will be compared. Following that, the results from the higher Reynolds number case ($Re = 405,000$) will be presented to show the effect of the Reynolds number on both the morphing wing and the trailing edge flap performance. Finally, the outcomes of the numerical investigations, selected based on the wind tunnel experimental results will be discussed.

A.1 Conventional Trailing Edge Flap and Ideal Trailing Edge Flap Performance

The conventional hinged trailing edge flap wing (Figure 5) and two ideal trailing edge wings (Figure 4) were designed to compare against the performance of the morphing wing. The ideal trailing edge flap does not possess the servos, servo horns, or the gap between the flap and the main wing as seen in the conventional flap cases. The

experimental lift coefficient (C_L) vs. angle of attack (α) for all three wings are shown in Figure 7a. Based on the McCormick formula [18], the lift curve slope ($dC_L/d\alpha$) for all wings at 0° flap deflection is $0.0545/\text{deg}$, and the corresponding effective AR is 2.80.

$$a = \frac{dC_L}{d\alpha} = a_0 \left(\frac{AR}{AR + 2 \left(\frac{AR + 4}{AR + 2} \right)} \right) \quad (1)$$

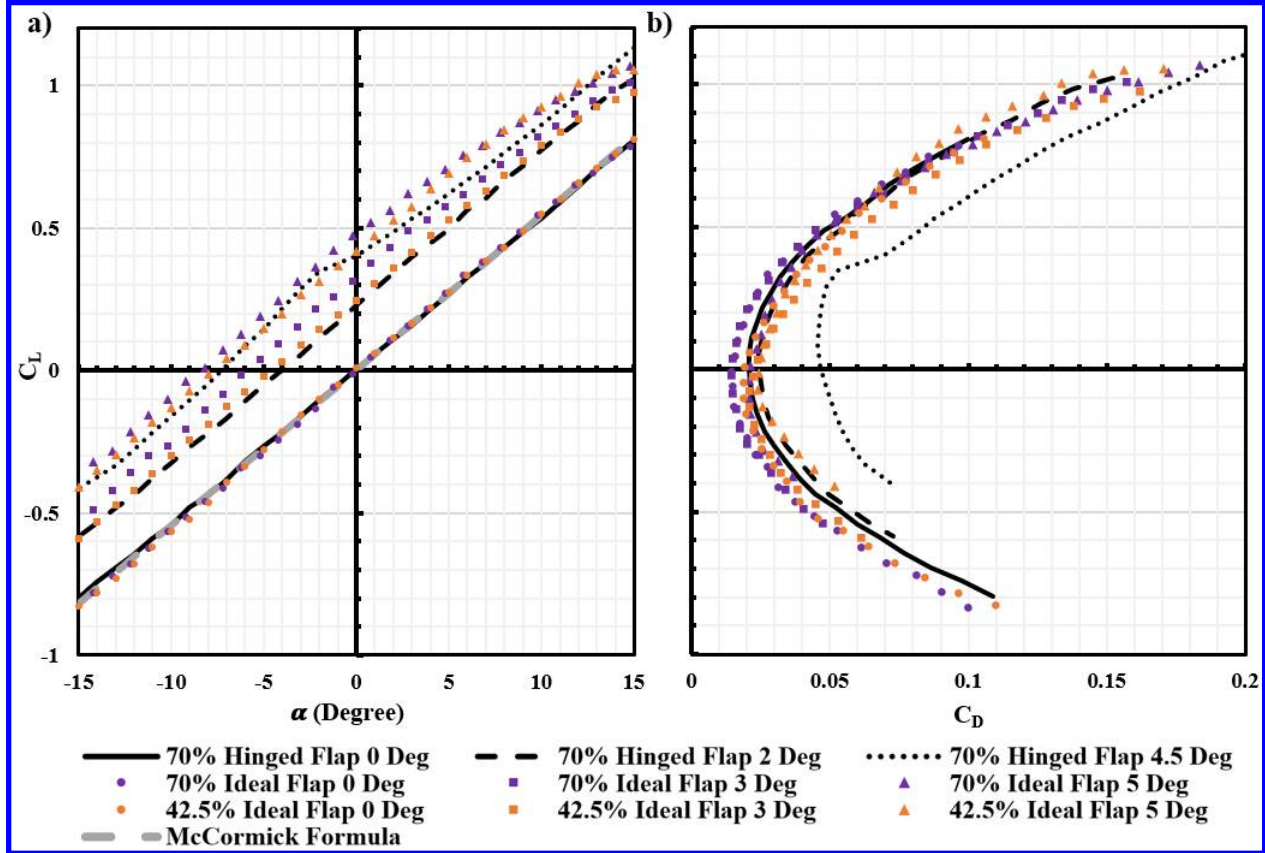


Figure 7. a) C_L vs. α and b) Drag Polar (C_L vs. C_D) for the Conventional and Ideal Trailing Edge Flap Wing under various δ_{eff} . ($Re = 270,000$)

Different colors in Figure 7 represent different c_{def} while different symbols represent different δ_{eff} . For the $\delta_{eff} = 0^\circ$ all three of the trailing edge device's performance overlaps as expected and aligns well with the theoretical result. As δ_{eff} is increased, C_L is increased at all α tested as expected. Comparatively, $0.7 c_{def}$ ideal flap sees the highest C_L increment while the $0.7 c_{def}$ hinged flap sees a lower increment. For $\delta_{eff} = 5^\circ$, a slight change in lift curve slope is observed for positive angles of attack when compared to negative angles of attack for hinged flaps.

Figure 7b shows the drag polar (C_L vs. C_D) for all cases is shown in Figure 7a. For $\delta_{eff} = 0^\circ$, the three wing configurations perform similarly however do not overlap. As δ_{eff} increases, the ideal flaps maintain their overall aerodynamic performance compared to $\delta_{eff} = 0^\circ$, while the hinged flap shows a slightly higher drag coefficient. As δ_{eff} is further increased, each wing sees a decrease in aerodynamic efficiency due to the increment in C_D at the same C_L . This performance loss is most notable for the hinged flap, as a significant increment in drag is observed when compared to all other cases. This is likely due to the flow separation at the hinge location, which will be explored later in the numerical result section later in the paper. In general, eliminating the gap between the leading edge and the trailing edge flap reduces the drag generated by the wing at large δ_{eff} . Additionally, increasing c_{def} decreases the overall C_D for conventional trailing edge device.

A.2 Morphing Wing with Various Chord Locations of Deflection (c_{def})

The experimental results of C_L for various morphing wing designs with different c_{def} is shown in Figure 8a. Each color corresponds to a morphing wing design with a specific c_{def} (Figure 3), and each symbol denotes a different δ_{eff} ranging from zero to full morphing deflection. At $\delta_{eff} = 0^\circ$, the performance of all four morphing wings is identical. As δ_{eff} increases, C_L also increases across all tested angles of attack and the results closely overlap. At their maximum δ_{eff} , each morphing wing reaches its max C_L . Except for the 0.425 c_{def} case, wings with a higher c_{def} tend to show a larger increase in C_L . It is also worth noting that $dC_L/d\alpha$ for all cases remains constant at $\alpha < 10^\circ$, which differs from the results for the conventional trailing edge flap results in Figure 7a, indicating the morphing wing has a more reliable lift generation capability at high δ_{eff} . It is additionally worth noting that the results for the 0.55 c_{def} is not shown in Figure 8 to improve clarity as its performance lies between 0.425 and 0.70 c_{def} cases.

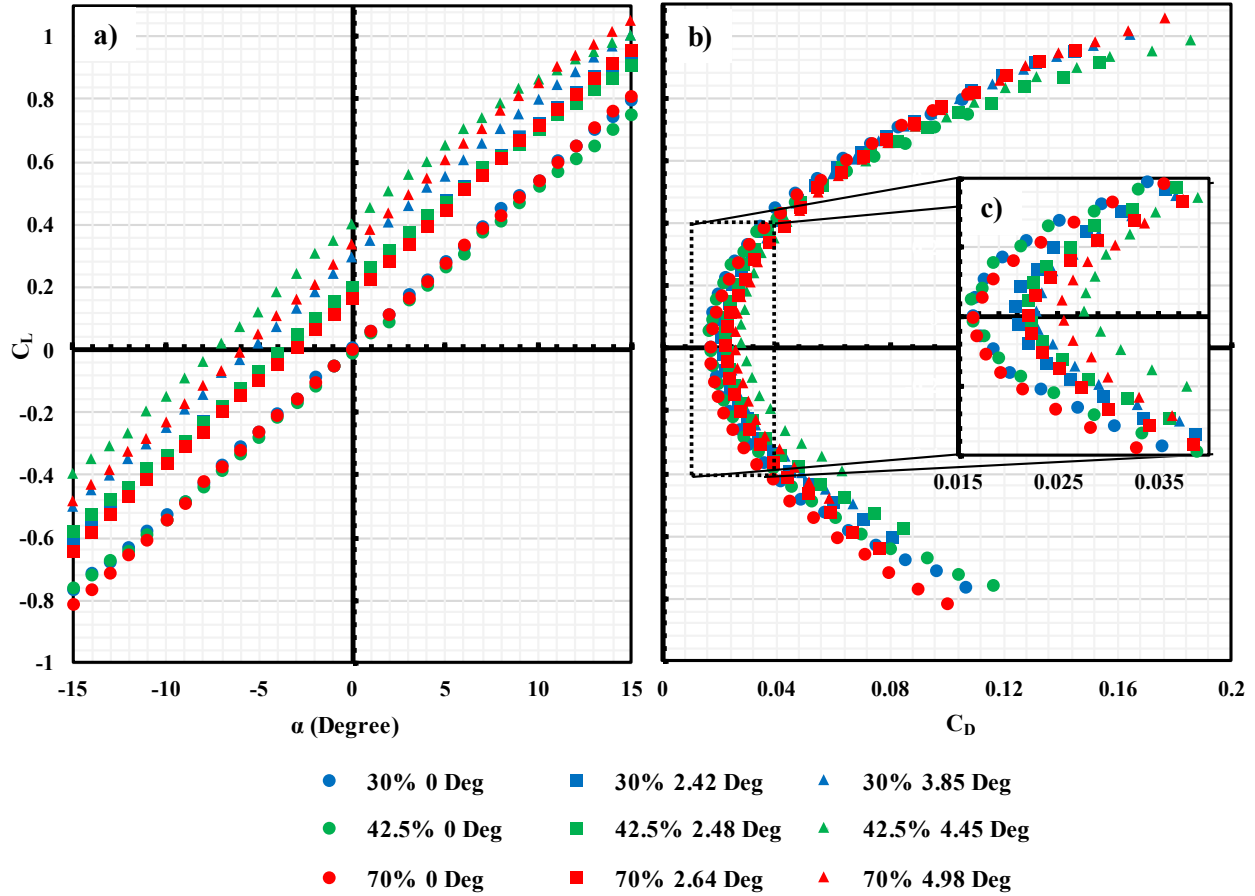


Figure 8. a) C_L vs. α , b) C_L vs. C_D and c) C_L vs. C_D at $C_L \sim 0$ for All Morphing Wing at Different Deflection Angles ($Re = 270,000$)

Figure 8b shows the drag polar for the same morphing wing cases in Figure 8a. At $\delta_{eff} = 0^\circ$, a small difference in C_{D0} is observed, attributed to differences in the wing's end cap design (Figure 2). To account for this small skin friction drag disparity, a correction to the drag coefficient (C_D) is applied. The average C_{D0} at $\delta_{eff} = 0^\circ$ is calculated, and the differences between each case's average at $\delta_{eff} = 0^\circ$ are determined, resulting in a correction ΔC_{D0} of up to 0.0016. This ΔC_{D0} is then applied to all cases. All $\delta_{eff} = 0^\circ$ cases, after correction, overlap and show a lower C_{D0} when compared to higher δ_{eff} cases, which contrasts with the results for the ideal flap case in **Error! Reference source not found.** In the range of $-0.4 < C_L < 0.4$, the aerodynamic performance of all c_{def} cases are nearly identical. At higher δ_{eff} , the 0.3 c_{def} morphing wing shows a lower C_D compared to other morphing wing designs. The 0.425 c_{def} case, which shows the highest C_L in Figure 8b, also generates the highest drag, indicating a reduced

overall performance. At $C_L > 0.4$, the aerodynamic performance for 0.3 and 0.7 c_{def} morphing wings are nearly identical for all δ_{eff} and outperform 0.425 and 0.55 c_{def} cases which shows a higher C_D at $C_L > 0.7$.

Figure 9 shows the drag polar for the two best-performing morphing wing designs (0.3 and 0.7 c_{def}), alongside the ideal and conventional hinged flap wing design at 0.7 c_{def} at their half and full deflections. Within the $-0.4 < C_L < 0.4$ range, both the morphing wing and the ideal flap exhibit a lower C_D compared to the hinged flap. This underscores the advantages of featuring continuous upper and lower surfaces as opposed to the discontinuous surfaces seen in morphing wings from literature using corrugated or sliding mechanisms [6-9]. At δ_{eff} of half deflection travel, a higher C_D is observed for both the morphing wing and hinged flap than the ideal flap. Further increasing δ_{eff} leads to an increment in C_D for all wings, with the conventional flap wing experiencing a particularly notable increase. At $C_L > 0.4$, the aerodynamic performance for the ideal flap and the morphing wing merges, becoming almost identical at $C_L > 0.55$. This similarity indicates their effectiveness in generating higher lift. Overall, the morphing wing demonstrates a performance very similar to that of the ideal flap design.

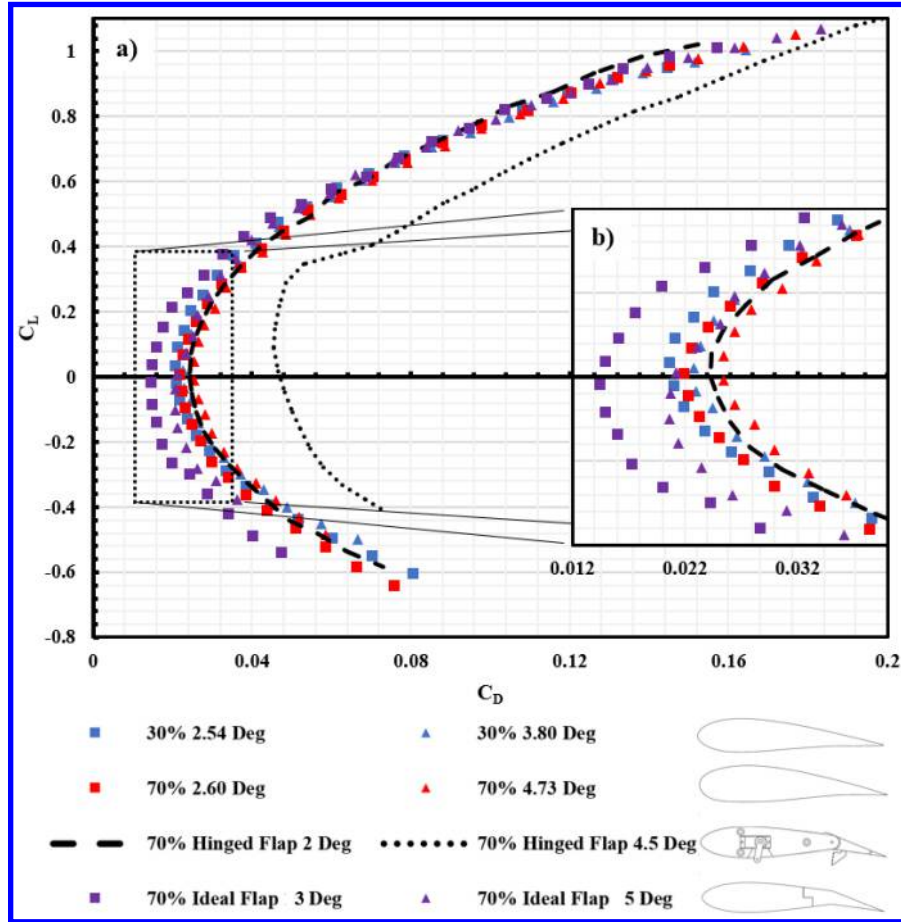


Figure 9. A) C_L vs. C_D and B) C_L vs. C_D at $C_L \sim 0$ for top-performing morphing wings and trailing edge wings ($Re = 270,000$)

A.3 Effect of Reynolds Number on Morphing Wing Performance

Several studies [12-14] highlight the significant benefits of higher Reynolds numbers on the aerodynamic performance of morphing wings. To examine the effect of Reynolds number, both the ideal flap and morphing wing at 0.7 c_{def} , which represent the best performers in their categories, were tested at a higher Reynolds number. It is important to note that the range of α tested in each case is constrained by the maximum load capacity of the ATI gamma sensor [15] used in the experiment.

Figure 10a shows the C_L vs α performance of the 0.7 c_{def} morphing wing (in blue) and the ideal flap (in red) at $Re = 405,000$. At $\delta_{eff} = 0^\circ$, the results for both cases overlap. As δ_{eff} is increased to servo half travel, both C_L and

the performance difference between the morphing wing and the ideal flap increases. Further increasing δ_{eff} results in a larger increment in C_L for the ideal flap when compared to the morphing wing. The morphing wing at a higher V_∞ and Reynolds number produced a lower C_L compared to its performance at a lower Reynolds number, as shown in Figure 9. It is suspected that the servo used for the morphing design was not able to generate enough torque to hold the morphing wing in shape at higher dynamic pressure thus decreasing the δ_{eff} during testing, while the ideal flap model is designed solid, and no servo was required for deflection.

Figure 10b shows the drag polar for the cases in Figure 10a. Again, at $\delta_{eff} = 0^\circ$, the two wings' performances overlap, validating the testing conditions. As δ_{eff} increases, the overall drag polar for the morphing remains relatively consistent, indicating stable performance across different δ_{eff} values. However, the ideal flap case sees a higher C_D increment for both $\delta_{eff} > 0$ cases. Compared to the lower Reynolds number shown in Figure 9, the morphing wing now sees a higher aerodynamic efficiency than the ideal flap, suggesting the advantages of the morphing wing design for higher Reynolds number applications.

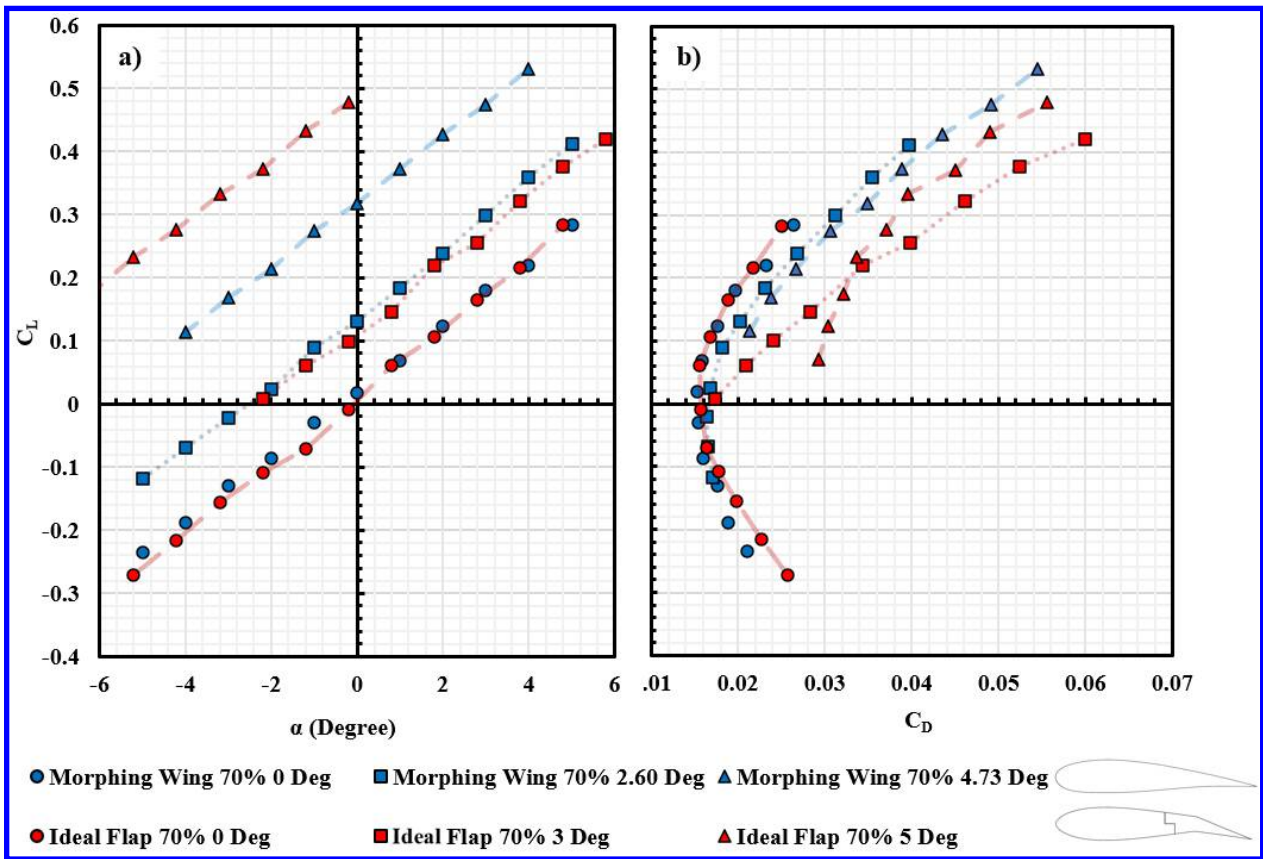


Figure 10. a) C_L and b) Drag Polar for 70% Morphing Wing and Ideal Flap ($Re = 405,000$)

B. Comparison of Experimental and Numerical Simulation (FlightStream®) Results

To further investigate the aerodynamic performance and the differences in pressure distribution between the morphing wing and the trailing edge flap design, FlightStream® simulations were conducted on the cases in Figure 9. The simulations were performed at maximum deflections and at $V_\infty = 20m/s$. Based on the results from our previous study [17], FlightStream® was found to accurately predict the C_L on the wing while tending to underpredict the drag force on the wing. Consequently, the simulation results for C_L were compared with the experimental data and are shown in Figure 11. The comparison revealed that, with the exception of the 0.7 c_{def} ideal flap case at $\alpha < 0^\circ$ (where FlightStream® overpredicted C_L), the simulation results from FlightStream® closely align with the experimental results. This consistency indicates that the pressure distribution results obtained from FlightStream® are reliable and can be used for further comparative analysis.

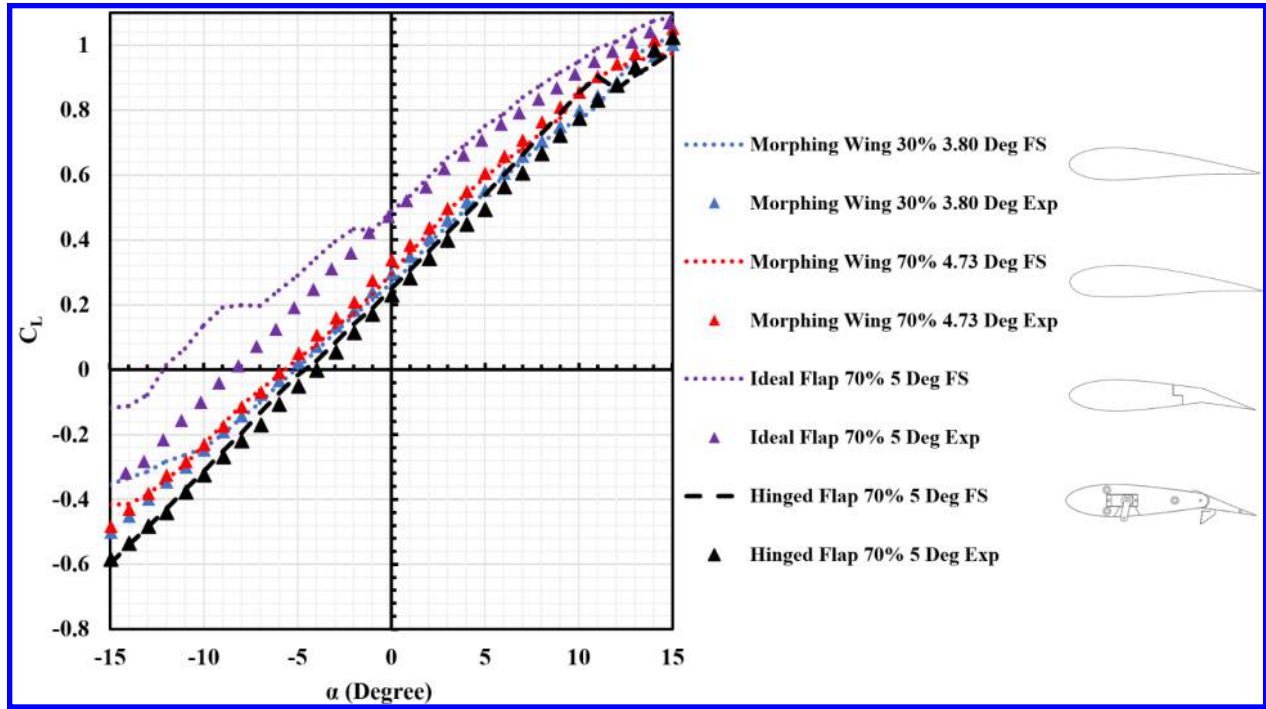


Figure 11. C_L vs α Comparison of FlightStream® and Experimental Results

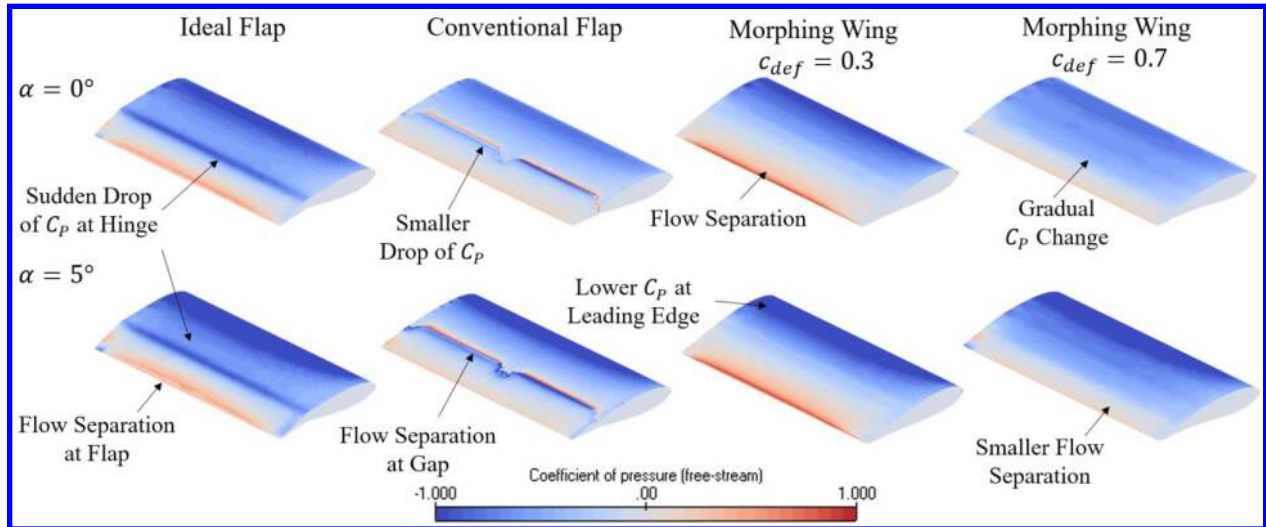


Figure 12. Pressure Coefficient (C_p) Contour for Cases in Figure 11 at a) $\alpha = 0^\circ$ and b) $\alpha = 5^\circ$

The pressure coefficient (C_p) contour calculated by FlightStream® at $\alpha = 0^\circ$ and 5° is shown in Figure 12. For the ideal flap wing, a sudden decrease in C_p is observed at the hinge location, attributed to the abrupt curvature in the area. Downstream of this section, flow separation is observed where C_p increases above 0.0. This results in a higher drag produced by the idea flap at $\delta_{eff} = 5^\circ$, as seen in Figure 9. For the conventional flap, a change in C_p is also observed at the hinge location, although the change of C_p magnitude is smaller than the ideal flap, resulting a less effective lift generation when deployed. Moreover, flow separation is also observed at the gap between the leading edge and trailing edge at higher α indicated by a relatively higher C_p . For the morphing wing with 0.3 c_{def} , a significant reduction in C_p (higher suction) occurs closer to the leading edge due to the morphing deflection location being nearer to this edge. However, flow separation is also observed closer to the trailing edge of the morphing wing due to the early increase in the airfoil curvature. Meanwhile, 0.7 c_{def} morphing wing has a much gradual change in C_p along the chord,

resulting a lower overall drag produced by the morphing wing, as seen in the results from both Figure 9 and Figure 10 while maintaining a similar lift performance.

IV. Conclusion

Several configurations of a novel Fishbone Skin-Actuated-Camber (SAC) morphing wings, along with ideal flaps and wings with conventional trailing edge flaps, were investigated experimentally and numerically. For the ideal flap design, removing the hinge and the gap between the leading edge and trailing edge (ideal flap) brings significant benefits in drag reduction, especially at higher flap deflection angles. However, it doesn't represent a realistic scenario. The hinged flap showed higher drag and similar lift performance as expected. The patent pending novel morphing wing design was able to achieve effective camber change while eliminating any buckling and gap on the surface of the wing and achieve a similar overall aerodynamic performance as the ideal flap configuration.

The chordwise location of deflection in the morphing wings had a relatively minor impact on the lift production when achieving a similar effective camber deflection angle. This is also confirmed by the FlightStream® numerical analysis, which agreed well with the experimental result. The simulated pressure contour results show no sudden discontinuity in pressure variation in the case of morphing wings as compared to the ideal and hinged flap cases. The simulation also provides evidence as to why $0.7 c_{def}$ shows greater overall performance when compared to other morphing wings. This is due to the reduced flow separation region on the upper surface of the wing at a positive angle of attack, resulting in an overall reduction in drag production.

V. Acknowledgment

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